

A survey of methods addressing sample selection bias

0. Introduction

Sample selection bias has always been a pervasive issue in gathering data (e.g., for medical treatments, econometrics, etc.) that attempts to be representative of the population. This issue becomes particularly serious when the data collected is analyzed and used for purposes like decision-making that negatively affects people's lives. The advent of big data has only amplified this problem. Biases in data collection (e.g., online surveys) propagate through data analysis and skew results in ways that are difficult to detect [1]. These biased results, if left undetected, can have significant implications for certain groups of people (e.g., minority groups) in a world that is becoming increasingly dependent on analytics driven by big data.

However, it is not only data and its collection process that can be biased; the statistical models themselves are also susceptible to being sources of bias that also yield potentially serious consequences for people. In particular, because addressing potential bias in increasingly large data sets is intractable, data analysis is instead devoted to building the model (e.g., choice of optimizer, number of hyperparameters, etc.) as a means of conflating a metric (e.g., test set accuracy) with bias [1]. As a result, there is often a lack of focus on the robustness and fairness of the model itself, which itself warrants more study [1]. In this report I present a survey of techniques, both data-driven and model-driven, that, to varying degrees of success, address sample selection bias.

1. Survey of methods

In this section, I present a number of methods (both old and recent) that all, in some way, address sample selection bias. I first discuss the Heckman model, linear regression, and extensions of the model. Then, I briefly review sample selection bias in classifier learning. Finally, I examine propensity scoring and recently proposed methods involving it.

1.1 Sample selection bias in linear regression

I begin by exploring extensions of and modifications to the Heckman selection model, which is an early method in correcting for sample selection bias affecting regression models in econometric applications.

1.1.1 Decomposing sample selection bias

Historically, many methods developed to address sample selection bias were not general solutions, for obvious reasons. They, instead, focused on either addressing sample selection bias

in a limited way or using mathematical tools to study sample selection bias theoretically and numerically [2]. In this subsection, I present a high-level overview of the latter approach to motivate some intuition behind the complexities of sample selection bias in regression models.

Stolzenberg and Relles [2] begin with explaining why the Heckman selection model, a method for estimating regression models with sample selection bias, gives unintuitive and difficult-to-interpret results. Simply put, this selection model is based on two equations Y_1 and Y_2 , where Y_1 is a regression equation of interest and Y_2 corrects for self-selection, called the selection equation. Simply put, the selection equation explicitly models the sampling probability of each observation. After a series of computations and transformations, the Heckman selection model reduces to the estimation of a multiple regression equation involving nonlinear transformations. While this approach may be valid, its results, though potentially significant, tend to be unintuitive and contradicting.

As a result, Stolzenberg and Relles derived an approximation to this method. They clarified and simplified the effect of selection bias by using Taylor series approximations and algebraic manipulations to derive the bias $\mathfrak{N}_{\beta_1}$ in the estimation of the regression coefficient β_1 that would be present if there were no correction for sample selection bias [2]. Further analysis showed that when this approach is combined with other selection bias correction tools, useful information is provided in helping to deal with selection bias. For a detailed proof as well as numerical simulations, see Stolzenberg and Relles' paper [2].

1.1.2 Extensions of Heckman's model

While Heckman's selection model provided a good benchmark for studying sample selection bias, its aforementioned incorporation of nonlinear transformations and generally unintuitive multiple regression equation warranted further study. In this subsection, I examine Vella's survey of extensions to the Heckman model [3]. He presents two main extensions: the first relaxes the distributional assumptions and the second changes the selection rules themselves.

Firstly, Vella addresses the primary criticism of parametric sample selection estimators of bias: their reliance on distributional assumptions (e.g., normality) [3]. His goal is to instead consider alternative estimators that rely less on parametric assumptions [3]. In particular, Vella employs index restrictions and semi-parametric methods that avoid defining the distributional assumptions of the model. However, this, of course, creates the obvious problems of losing the

tools of having explicitly defined the distributional assumptions. Secondly, Vella addresses the selection rules defined by the selection equation and expands the estimator to take more than just values of zero or one [3]. Depending on the application, he adjusts the selection rules (a.k.a. censoring rules) for multiple, ordered, and unordered indices. More details on both approaches are found in his paper [3].

1.2 Sample selection bias in classifier learning

As machine learning grows in popularity, so does sample selection bias in trained models. Here, I examine this bias's effect on classifier learning, how this bias breaks the principle assumption that all of the training data come from the same distribution of test data, and a bias correction method proposed by Zadrozny [4]. In defining sample selection bias in classifier learning algorithms, Zadrozny notes that if the sample selection s is independent of its label y given the feature vector x , the selected sample is biased and its level of bias depends on x . He argues this should be of concern because the primary interest of classifier learning is in the model's predictive accuracy [4]. To study this problem in machine learning, Zadrozny examines the effects of sample selection bias on Bayesian classifiers, logistic regression, decision tree classifiers, and support vector machines (SVM).

Firstly, Bayesian classifiers are unaffected by sample selection bias; when using the biased samples as training data in Bayes' rule, applying Bayes' rule again simply decomposes the expression to the posterior probability [4]. On the other hand, naive Bayes learners are affected by this bias. Secondly, logistic regression is also unaffected by sample selection bias, as the assumption that y is independent of s given x merely decreases the number of examples. However, in the asymptotic limit of the number of training examples, the results on any selected sample approach the results on a random sample [4]. Thirdly, decision trees are affected by sample selection bias because the splitting criteria are sensitive to the bias [4]. Lastly, hard margin SVM is not affected by sample selection bias because on separable data, sample size does not affect the separator in any significant way [4]. However, soft margin SVM is affected by the bias as varying densities of the feature space (X) changes the decision boundary.

To address this effect in classifier learning, Zadrozny showed that, under the assumption that s and y are independent given x , the expected value of the loss function under sample selection bias [4]. Beyond this, as discussed earlier, he found that different types of learners were affected by sample selection bias, which in turn defined a new categorization of classifiers.

1.3 Propensity scoring

Another method is called propensity scoring, proposed by Rosenbaum and Rubin in 1983 [5]. Specifically, this method was studied in the context of observational studies and causal effects (e.g., medical trials). The propensity score is defined as “the conditional probability of assignment to a particular treatment given a vector of observed covariates” [5]. Rosenbaum and Rubin also show that propensity score adjustment is sufficient for removing the effects of bias for both small and large samples. Furthermore, many techniques for propensity score matching (PSM), a method balancing the propensity scores of the treatment and comparison groups to ensure meaningful analysis, have been developed to mitigate the effects of selection bias. For more details, please see Bai’s article comparing various PSM methods [6].

1.4 State-of-the-art approaches

I now present two recently proposed methods that tackle the prominence of the internet and big data and move beyond propensity scoring. The first combines propensity scoring with another technique called statistical matching [7]. The second is a target-independent inference approach [8] that is competitive with propensity scoring.

1.4.1 Statistical matching and propensity score adjustment

In recent times, online surveys have grown rapidly in popularity as a method of data collection due to their convenience. This method, however, has two notable shortcomings: they rely on self selection and internet coverage. Consequently, these weaknesses make the method non-probabilistic and produce biased samples. Some methods, such as statistical matching and propensity score adjustment (PSA), have been developed to correct for the bias. Statistical matching is a machine learning method that trains a model that predicts values that can be useful to estimate population values. PSA trains a model that estimates propensity scores for participation in the sample. In this subsection, I present a brief overview of several methods that combine both statistical matching and PSA, as proposed by Castro-Martin, et al [7].

Firstly, shrinkage was employed in the development of an estimator based on composite information. That is, the estimator is a sum of the statistical matching estimator and the PSA estimator, each multiplied by constants that sum to one [7]. Here, shrinkage is a natural choice in improving estimates in terms of the mean squared error (MSE). Secondly, Castro-Martin, et al propose an estimator defined by a difference estimator [7]. Put simply, this method assesses the differences in a treatment group before and after the treatment is administered. Finally, they

propose computing PSA weights using some machine learning classifier. A statistical model (e.g., linear regression) is then trained using the weights and the goal is to minimize the weighted MSE [7]. For more details, including numerical simulations, see [7].

1.4.2 Universal adaptability

Finally, I end by discussing a recent method developed by Kim, et al that is proposed as a competitive alternative to propensity scoring [8]. As has been discussed, drawing statistically valid results from data involve truly random sampling from the population. However, as mentioned, this is a difficult task, which is why methods such as propensity scoring have been developed. Here, Kim, et al present a method that they dub “universal adaptability”, which makes inferences based on available data from a source population that allows for good estimates for any test data [8]. This strategy is based on imputation, which is a process consisting of “learning a prediction function that estimates the relationship of the variable of interest given the covariates of the individual” [8] and “averaging [that] prediction over the target distribution” [8]. Furthermore, their numerical simulations show this method is effective in multiple contexts. For more, including theoretical guarantees of this approach, see [8].

2. Conclusion

What can be drawn from our discussion is the wide range of perspectives from which one may try to address the effects of sample selection bias. This bias is one that has been extensively studied, but is one that I believe is even more relevant in the world today. The advent of big data contributed to the rapid growth of data-driven decision-making, which consequently leads to increasing numbers of people impacted by these analytics. While this tool can very much be of great benefit, biases such as sample selection bias still affect the data collection process and lead to biases propagating through algorithms run on the data and skews the results. However, the rigorous work done by those mentioned in this report, along with many others who have contributed to developing methods in this topic, have given statisticians and data scientists clearer insights into ways to mitigate sample selection bias. I believe this is still a field of active interest, and being able to answer questions in this field will help push us one step closer towards algorithmic fairness.

References

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